

Monte Carlo Pricing of Mortgage-Backed Securities with Stochastic Prepayment and Default Risk

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Abstract

This paper develops a Monte Carlo framework for pricing collateralized mortgage obligations (CMOs) under stochastic interest rates. Interest rates are modeled using the Cox-Ingersoll-Ross (CIR) process, and mortgage cash flows are generated from a standard amortization structure. The analysis begins with a baseline model assuming constant prepayment, and is then extended to incorporate stochastic prepayment and default behavior driven by interest rate dynamics. Numerical results show that stochastic prepayment reduces CMO values by accelerating the return of principal, while default risk has a larger negative impact due to direct cash flow losses. Sensitivity analysis further demonstrates that interest rates, default intensity, and mortgage rates are the primary drivers of valuation. Overall, the results highlight the importance of modeling borrower behavior and credit risk in the valuation of mortgage-backed securities.

1 Introduction

Mortgage-backed securities (MBS) are a major component of modern financial markets, representing trillions of dollars in outstanding debt. In the United States alone, there is more than \$11 trillion of securities outstanding and close to \$300 billion in average daily trading volume, making the MBS market one of the largest and most liquid fixed-income markets globally [5]. These securities allow lenders to pool mortgages and transfer risk to investors. Their valuation is significantly more complex than that of traditional bonds due to the embedded options held by borrowers.

The cash flows of mortgage-backed securities are highly uncertain. Borrowers have the ability to prepay their loans, often in response to changes in interest rates, and may also default if economic conditions worsen. These features introduce both timing and credit risk into the cash flow structure. As a result, the value of an MBS depends on both the market interest rate and how borrowers respond to changing economic incentives. Because of these complexities, analytical pricing formulas are generally not available, and numerical methods are required. Monte Carlo simulation provides a natural framework for this problem, as it allows for the modeling of stochastic interest rates and path-dependent cash flows. In particular, it enables the incorporation of behavioral features such as prepayment and default that depend on how interest rates change over time.

In this paper, we study the pricing of a simplified collateralized mortgage obligation (CMO) using Monte Carlo methods and stochastic interest rates modeled by the Cox-Ingersoll-Ross (CIR) process. We begin with a baseline model that assumes a constant prepayment rate, and then extend the framework to include stochastic prepayment and default rates that depend on the difference between the mortgage rate and the market interest rate. These extensions reflect the economic intuition that borrowers refinance when rates fall and are more likely to default when rates rise. This paper attempts to understand how these modeling choices affect both the valuation and behavior of mortgage-backed securities. By comparing the base model with increasingly realistic extensions and performing sensitivity analysis, we highlight the importance of interest rates, prepayment behavior, and default risk in determining CMO prices.

2 Mortgage and CMO Structure

Mortgage-backed securities are constructed from pools of residential mortgage loans, where investors receive cash flows generated by homeowners' monthly payments. To model and price these securities, it is essential to first understand the structure of the underlying mortgage and the way cash flows are generated and distributed.

2.1 Mortgage Structure

Consider a mortgage with loan amount L , fixed monthly interest rate m , and duration of D months. Borrowers make fixed monthly payments designed to fully amortize the loan over

its lifetime. The payment amount is given by

$$P = \frac{mL}{1 - (1 + m)^{-D}}.$$

This standard amortization formula is widely used in mortgage modeling and other fixed-income scenarios [6]. Each monthly payment consists of both interest and principal repayment. At month t , the interest portion is calculated as the product of m and the remaining principal at the beginning of the month. The remainder of the payment is applied toward reducing the principal balance. Over time, the composition of the payment shifts: interest makes up a substantial percentage of early payments, while later payments consist mostly of principal. This amortization structure plays an important role in determining the timing of cash flows received by investors. In addition to scheduled payments, mortgage cash flows may include unscheduled reductions in principal. In this model, the total cash flow in a given month consists of three components: the interest payment, scheduled principal repayment, and prepayment. The total principal reduction in a given month is therefore the sum of scheduled principal and any prepayment. After these payments are made, the remaining balance is updated and carried forward to the next period. These cash flows continue until the mortgage is fully paid off, either through scheduled amortization or early termination.

Prepayment refers to the early repayment of part or all of the outstanding mortgage balance. This can occur for a variety of reasons, including refinancing, moving, or other borrower-specific factors. In the base model, prepayment is modeled as a constant fraction of the remaining balance each month. Under this assumption, a fixed proportion of the outstanding principal is prepaid at every time step. While this provides a baseline, it does not capture the known dependence of prepayment behavior on economic conditions, particularly interest rates. More realistic models treat prepayment as a dynamic process influenced by refinancing incentives. This perspective is supported by the mortgage option frameworks developed in [2] and [3]. This idea will be developed in Section 5, where prepayment rates are allowed to vary with the simulated interest rate path.

In addition to prepayment, mortgages are subject to default risk. Default occurs when borrowers fail to meet their payment obligations, leading to a loss of future cash flows for investors. In the base model, default is not included. However, in practice, default risk represents a key source of uncertainty in mortgage-backed securities. When defaults occur, investors may recover only a fraction of the outstanding principal, depending on recovery assumptions and market conditions. As with prepayment, a stochastic intensity model depending on interest rates will be developed later, in Section 6.

2.2 Collateralized Mortgage Obligations

A collateralized mortgage obligation (CMO) is a structured financial product backed by a pool of mortgages. The total cash flows generated by the mortgage pool are redistributed to investors according to a predefined structure. In general, CMOs are divided into tranches, each with different payment priorities and risk characteristics. Higher-priority tranches receive payments first and are therefore less exposed to prepayment and default risk, while lower-priority tranches face greater risk but offer higher potential returns [6]. To isolate

the core effects of interest rates, prepayment, and default, this paper considers a simplified structure consisting of a single tranche that receives all cash flows from the mortgage pool. This simplification allows the analysis to focus directly on how underlying mortgage behavior affects valuation, without the additional complexity of tranche interactions.

Under the single-tranche assumption, the investor receives all mortgage cash flows, including interest payments, scheduled principal repayments, and prepayments, while also experiencing losses due to default. These cash flows occur monthly and depend on the evolution of the remaining mortgage balance over time. The value of the CMO is therefore determined by the present value of this stream of uncertain cash flows. In this paper, future cash flows are simulated under stochastic interest rates and then discounted back to the present. The interaction between interest rates and mortgage behavior, particularly through prepayment and default, plays a key role in determining both the timing and magnitude of these cash flows.

3 Interest Rate Model

The valuation of mortgage-backed securities depends on the evolution of interest rates over time. Interest rates influence both the discounting of future cash flows and borrower behavior, specifically through prepayment and default decisions. To capture these effects, a stochastic interest rate model is required.

3.1 The Cox-Ingersoll-Ross (CIR) Model

In this paper, interest rates are modeled using the Cox–Ingersoll–Ross (CIR) model. The interest rate $r(t)$ follows the stochastic differential equation

$$dr(t) = \alpha(b - r(t))dt + \sigma\sqrt{r(t)}dW(t),$$

where:

- $\alpha > 0$ is the speed of mean reversion
- $b > 0$ is the long-run mean level
- $\sigma > 0$ controls volatility
- $W(t)$ is a standard Brownian motion.

The CIR model is a mean-reverting process. When the interest rate is above the long-run level b , the drift term becomes negative and pulls the rate downward, while rates below b are pushed upward. This behavior is consistent with empirical observations of interest rate dynamics. A key feature of the CIR model is that it preserves non-negativity of interest rates. Unlike models with constant volatility, such as the Vasicek model, the diffusion term $\sigma\sqrt{r(t)}$ decreases as $r(t)$ approaches zero, preventing the process from becoming negative under appropriate parameter conditions ($2\alpha b \geq \sigma^2$) [6].

3.2 Simulation of Interest Rate Paths

Although the CIR process does not admit a simple closed-form solution for $r(t)$, its conditional distribution is known. Given $r(s)$, the value of $r(t)$ for $t > s$ follows a scaled noncentral chi-squared distribution. Specifically, the transition from $r(t_{i-1})$ to $r(t_i)$ can be written as

$$r(t_i) = c_i x_i,$$

with x_i being a sample generated from a noncentral $\chi^2(k, \lambda_i)$, where

$$k = \frac{4b\alpha}{\sigma^2}, \quad \lambda_i = \frac{4\alpha e^{-\alpha(t_i - t_{i-1})}}{\sigma^2(1 - e^{-\alpha(t_i - t_{i-1})})} r(t_{i-1}), \quad c_i = \frac{\sigma^2(1 - e^{-\alpha(t_i - t_{i-1})})}{4\alpha}.$$

This property allows the CIR process to be simulated exactly at discrete time points [6].

To generate interest rate paths, the time interval $[0, T]$ is discretized into equally spaced steps corresponding to the monthly frequency of mortgage cash flows. Starting from an initial rate $r(0)$, the process is simulated recursively using the conditional distribution described above. At each step, $x_i \sim \chi^2(k, \lambda_i)$ is generated and the interest rate is obtained by computing $r(t_i) = c_i x_i$. The process is repeated for the full time horizon. This produces a path $r(t_0), r(t_1), \dots, r(t_D)$, where each step corresponds to one month. In the implementation, interest rates are simulated on an annualized basis and then converted to monthly rates in order to discount the cash flows.

The CIR model is chosen over alternative short-rate models, such as the Vasicek model, for several reasons. First, the CIR model ensures that interest rates remain non-negative, which is important in the context of discounting and mortgage valuation. Second, the state-dependent volatility in the CIR model provides a more realistic representation of interest rate dynamics. When rates are low, volatility decreases, reflecting reduced variability near zero. Finally, the fact that its conditional distribution is known allows for efficient generation of interest rate paths, making the CIR model well-suited for Monte Carlo simulation.

4 Base CMO Pricing Model

This section describes the baseline model used to price the collateralized mortgage obligation. The framework follows the Monte Carlo approach presented by Ökten in [6], with a modification to the discounting procedure that more accurately reflects the time evolution of interest rates.

4.1 Mortgage Cash Flow Generation

The model considers a mortgage pool with initial principal L , fixed monthly mortgage rate m , and duration of D months. Borrowers make fixed monthly payments, and in each period the payment is divided into interest and scheduled principal repayment. In the base model, prepayment is assumed to occur at a constant rate throughout the life of the mortgage. This implies that the same fraction of the outstanding balance is prepaid in every period, independent of interest rates or other economic conditions. The total principal reduction

is therefore given by: total principal = scheduled principal + prepayment. To make sure that the remaining balance never becomes negative, the total principal payment is capped at the outstanding balance. The life of the mortgage ends once the balance reaches zero. The cash flow received by the investor in each month consists of the interest payment and the total principal payment. Under the constant prepayment assumption, the evolution of the remaining balance, and therefore the entire sequence of cash flows, is deterministic. In particular, the time at which the mortgage is fully paid off is deterministic, and does not vary from one simulation to another. Although this assumption is not realistic, it provides a useful benchmark. It allows the model to isolate the effect of stochastic interest rates on valuation, without introducing additional sources of randomness from borrower behavior. This baseline will serve as a point of comparison for the extended models developed in later sections.

4.2 Monte Carlo Pricing Framework

The price of the CMO is computed as the present value of future monthly cash flows. Since discounting depends on the evolution of interest rates, the expectation is approximated using Monte Carlo simulation. For each simulation, an interest rate path is generated using the CIR model described in Section 3, the (deterministic) mortgage cash flows are computed, and then each cash flow is discounted to the present using the simulated interest rates. This procedure is repeated across N independent paths, and the CMO price is estimated as the average of the discounted cash flows across all simulations. In this base model, variation across simulation paths comes solely from differences in the discount factors, since the underlying cash flows are identical in every path.

A key component of the model is the method used to discount future cash flows. In the original implementation in [6], discounting is performed using powers of the simulated interest rates, which implicitly assumes that each rate applies uniformly over all prior periods. In this paper, a path-wise discounting approach is used instead. The discount factor is updated recursively along each simulated path according to

$$\text{DF}_n = \prod_{i=1}^n \frac{1}{1 + r_i},$$

where r_i is the simulated monthly interest rate at time i . Each cash flow is then multiplied by the corresponding discount factor. This approach ensures that each period's interest rate only affects discounting going forward, providing a more accurate representation of the time evolution of rates. The modification is straightforward to implement and does not significantly increase computational cost.

4.3 Algorithm Description

The full pricing algorithm can be summarized as follows:

1. Initialize model parameters and compute the fixed monthly mortgage payment
2. For each of the N simulation paths:

- Initialize the remaining mortgage balance
- Generate a path of interest rates using the CIR model and convert to monthly values
- Iterate through each month:
 - Compute interest and scheduled principal
 - Apply prepayment
 - Compute total cash flow
 - Update the discount factor recursively
 - Accumulate discounted cash flows
 - Update the remaining balance

3. Compute the Monte Carlo estimate as the average across all paths

In addition to the price, the simulation records the effective mortgage life, measured as the number of months until the balance is fully paid off. Under the constant prepayment assumption, this quantity is the same across all simulation paths and serves as a useful summary of the timing of cash flows in the base model.

5 Stochastic Prepayment Model

The constant prepayment assumption in the base model provides a useful benchmark, but it fails to capture one of the most important features of mortgage-backed securities: the dependence of prepayment behavior on interest rates. In reality, prepayment activity is highly sensitive to refinancing incentives, which vary over time with market conditions. This section extends the base model by introducing a stochastic prepayment mechanism that depends on the simulated path of interest rates.

5.1 Economic Intuition

Prepayment behavior is closely tied to the difference between the mortgage rate and market interest rates. When market rates fall below the mortgage rate, borrowers have an incentive to refinance their loans at a lower rate, leading to increased prepayment. Conversely, when market rates are above the mortgage rate, refinancing is unattractive and prepayment activity tends to decrease.

This relationship is often interpreted through the lens of an embedded option. The borrower effectively holds a call option on the mortgage, which can be exercised by refinancing when it becomes advantageous. This option-theoretic view of mortgage behavior is well established in the literature, including in [1] and [4]. Empirical evidence further supports the strong dependence of prepayment rates on interest rate movements (see [2] and [7]).

5.2 Model Specification

To incorporate this behavior, the prepayment rate is modeled as a time-varying function of the interest rate path. At each time t , the prepayment rate is defined as

$$\lambda_p(t) = a_p \exp(b_p(m - r(t))^+),$$

where:

- $a_p > 0$ is a baseline prepayment level
- $b_p > 0$ controls the sensitivity to refinancing incentives
- m is the fixed mortgage rate
- $r(t)$ is the current market interest rate
- $(x)^+ = \max(x, 0)$.

This specification captures two key features. First, prepayment increases when $r(t) < m$, reflecting refinancing incentives. Second, the exponential form allows prepayment rates to rise rapidly as the incentive grows, consistent with observed borrower behavior [3]. The inclusion of the baseline parameter a_p ensures that prepayment can occur even in the absence of a rate incentive, reflecting exogenous factors such as relocation or changes in borrower circumstances.

5.3 Implementation in the Monte Carlo Framework

The introduction of stochastic prepayment modifies only the cash flow generation step of the base model. All other components, including interest rate simulation and discounting, remain unchanged. At each time step along a simulated interest rate path, the prepayment rate is computed as described in the previous section, using the current value of $r(t)$. The prepayment amount is then calculated as the corresponding fraction of the remaining balance, replacing the constant prepayment assumption used in the base model.

This modification has an important consequence. The evolution of the mortgage balance is no longer deterministic. Since prepayment rates now depend on the simulated interest rate path, both the timing and magnitude of cash flows vary across simulations. In particular, paths with lower interest rates tend to produce higher prepayment rates, leading to faster repayment of principal and a shorter effective mortgage life. Conversely, higher-rate paths result in slower prepayment and a longer mortgage life. As a result, the Monte Carlo simulation now captures both discount rate uncertainty and variability in borrower decisions. In addition to the CMO price, the simulation records the average prepayment rate and the average mortgage life across paths, providing a quantitative measure of how interest rate dynamics influence borrower behavior.

6 Default Risk Model

In addition to prepayment, mortgage-backed securities are exposed to default risk. While prepayment accelerates the return of principal, default reduces the total cash flows received by investors. This section extends the model by incorporating stochastic default behavior alongside stochastic prepayment.

6.1 Economic Intuition

Default behavior is also influenced by the relationship between mortgage rates and market conditions. When interest rates rise above the mortgage rate, borrowers may face increased financial stress or reduced refinancing opportunities, leading to a higher likelihood of default. Conversely, when rates are low, borrowers are less likely to default and more likely to refinance instead. As with prepayment, default can be interpreted within an option-theoretic framework, where borrowers choose between continuing payments, refinancing, or defaulting depending on economic incentives [3].

6.2 Model Specification

Similarly to prepayment, the default rate is modeled as a time-varying function of the interest rate path. At each time t , the default rate is defined as

$$\lambda_d(t) = a_d \exp(b_d(r(t) - m)^+),$$

where:

- $a_d > 0$ is a baseline default level
- $b_d > 0$ controls sensitivity to the default incentive
- m is the fixed mortgage rate
- $r(t)$ is the current market interest rate.

This specification captures the idea that default risk increases when market rates exceed the mortgage rate, and remains relatively low otherwise. The exponential form allows default rates to increase rapidly in high-rate environments, reflecting worsening borrower conditions. The baseline parameter ensures that default can occur even when rates are low, due to exogenous borrower-specific factors.

6.3 Implementation and Cash Flow Impact

The introduction of default modifies the cash flow dynamics in a fundamental way. At each time step, the total reduction in the mortgage balance now consists of three components:

$$\text{principal reduction} = \text{scheduled principal} + \text{prepayment} + \text{default}.$$

However, unlike prepayment, default does not generate a corresponding cash flow to the investor. In the model, defaulted principal is treated as a loss, meaning that only scheduled

principal and prepayments contribute to investor cash flows. As in the stochastic prepayment model, the default rate is computed at each time step based on the simulated interest rate path. The resulting default amount reduces the remaining balance and affects future cash flows.

This extension introduces an additional source of variability across simulation paths. While prepayment primarily affects the timing of cash flows, default reduces their overall magnitude. High interest rate paths tend to generate higher default rates, reducing total cash flows through principal losses. At the same time, lower prepayment rates in these environments can lead to longer mortgage lives, with the overall timing determined by the interaction of these effects. The Monte Carlo simulation tracks the average default rate and mortgage life across paths, providing insight into how default risk interacts with prepayment and interest rate dynamics in determining CMO valuation.

7 Full Monte Carlo Pricing Algorithm

The full pricing model combines the three main components developed in the previous sections: stochastic interest rates generated by the CIR model, interest-rate-dependent prepayment, and interest-rate-dependent default. For each simulated path, the model produces a sequence of monthly mortgage cash flows and discounts them back to time zero. The CMO price is then estimated as the average present value across all simulated paths. Relative to the base model, the main change is that cash flow generation is now path-dependent. Since both prepayment and default rates depend on the simulated interest rate path, the remaining mortgage balance evolves differently across simulations. As a result, both the timing and magnitude of cash flows vary from path to path.

At each month n , the model first computes the scheduled mortgage payment components. It then evaluates the prepayment and default rates using the current simulated interest rate. These are used to determine the prepayment amount and the default amount for that month. The balance is reduced by scheduled principal, prepayment, and default, while investor cash flow includes interest, scheduled principal, and prepayment, but excludes defaulted principal. Each monthly cash flow is discounted recursively using the simulated path of monthly interest rates. The full Monte Carlo estimate is therefore obtained by repeating this procedure over N independent interest rate paths and averaging the resulting present values. In addition to the estimated CMO price, the simulation records summary quantities such as the average prepayment rate, average default rate, and average mortgage life.

7.1 Algorithm Description

The full pricing algorithm can be summarized as follows:

1. Specify the mortgage parameters, prepayment parameters, default parameters, and CIR interest rate parameters
2. Compute the fixed monthly mortgage payment
3. For each of the N simulation paths:

- Initialize the remaining mortgage balance
 - Simulate a monthly interest rate path using the CIR model
 - Set the path discount factor equal to one
 - For each month until maturity or full payoff:
 - Compute interest and scheduled principal
 - Compute the prepayment rate from the current interest rate
 - Compute the default rate from the current interest rate
 - Calculate prepayment and default amounts
 - Update the remaining balance
 - Calculate the investor cash flow
 - Update the discount factor recursively
 - Add the discounted cash flow to the path present value
4. Average the path present values across all simulations to obtain the CMO price estimate.

8 Numerical Results

This section presents the results of the Monte Carlo simulations for the base model, the stochastic prepayment model, and the full model incorporating both stochastic prepayment and default. All results are based on $N = 10,000$ simulated interest rate paths under the CIR model. The goal is to understand how increasingly realistic modeling assumptions affect both the level and structure of CMO valuation, and to quantify the sensitivity of the model to key parameters.

All simulations and computations were implemented in Julia within a Jupyter Notebook environment. The complete code is provided in the GitHub repository. Table 1 shows baseline model parameters that were used throughout the implementation. Baseline mortgage and interest rate parameters are chosen consistent with the example by Ökten in [6], including a 30-year mortgage duration, a \$10 million mortgage pool, and CIR interest rate dynamics. While the functional forms of prepayment and default intensities are motivated by the literature, the specific parameter values are selected to produce realistic baseline behavior and stable Monte Carlo estimates.

Category	Parameter	Description	Value
Mortgage			
	L	Initial loan balance	10,000,000
	m	Mortgage rate (monthly)	0.065/12
	D	Duration (months)	360
Interest Rate (CIR)			
	r_0	Initial interest rate	0.045
	b	Long-run mean rate	0.05
	α	Mean reversion speed	0.3
	σ	Interest rate volatility	0.03
Prepayment			
	a_p	Base prepayment intensity	0.002
	b_p	Incentive sensitivity	80.0
Default			
	a_d	Base default intensity	0.0005
	b_d	Incentive sensitivity	100.0
Simulation			
	N	Number of Monte Carlo paths	10,000

Table 1: Baseline Model Parameters

8.1 Model Comparison

We begin by comparing the three models developed in the previous sections: the base model with constant prepayment, the stochastic prepayment model, and the full model with both stochastic prepayment and default.

Model	Price	Avg Life (months)	Prepayment	Default
Base	\$11.36M	229.0	0.00200	–
Stochastic Prepay	\$11.31M	220.6	0.00223	–
Prepay + Default	\$10.80M	206.4	0.00224	0.0005006

Table 2: Comparison of CMO prices across models.

Notes: Parameters used are those in Table 1.

As shown in Table 2, introducing stochastic prepayment reduces the CMO price relative to the base model. This reflects the fact that the model accounts for accelerating prepayment when interest rates fall, causing investors to lose higher-interest future cash flows. This also shortens the average life of the mortgage, as expected. The addition of default risk leads to a further and more substantial decrease in price. Unlike prepayment, which primarily alters the timing of cash flows, default results in a direct loss of principal and future income, making it a more severe source of risk. These results highlight the importance of incorporating both behavioral and credit risk in mortgage-backed securities valuation.

8.2 Prepayment Effects

We next examine the impact of prepayment behavior on CMO valuation by varying the parameters of the stochastic prepayment model.

Scenario	Price	Avg Life (months)	Prepayment Rate
Low Sensitivity	\$11.34M	224.6	0.00211
Moderate	\$11.31M	220.6	0.00223
High Sensitivity	\$11.27M	213.4	0.00247

Table 3: Sensitivity to prepayment intensity

Notes: Parameters used are those in Table 1, apart from b_p which is varied between 40.0 (low), 80.0 (moderate), and 150.0 (high).

As shown in Table 3, increasing the sensitivity of prepayment to interest rates leads to higher prepayment rates, shorter mortgage lifetimes, and lower CMO prices. This goes along with what we would expect to see for mortgage-backed securities. When borrowers recognize more incentive to refinance, prepayments accelerate, reducing the value of future high-interest payments.

Baseline Prepayment	Price	Avg Life (months)
0.002	\$11.31M	220.6
0.004	\$11.03M	171.6

Table 4: Impact of baseline prepayment

Notes: Parameters used are those in Table 1, apart from a_p which is varied between 0.002 and 0.004.

As shown in Table 4, increasing the baseline prepayment level has a particularly strong effect on valuation. Higher baseline prepayment leads to consistently faster principal repayment, reducing the overall amount of interest collected and significantly lowering the CMO price. This indicates that the baseline prepayment level plays a much larger role in determining the CMO price than the sensitivity to the prepayment incentive.

8.3 Default Effects

We now examine the impact of introducing default risk into the model.

Scenario	Price	Avg Life (months)	Default Rate
Moderate Sensitivity	\$10.803M	206.4	0.0005006
High Sensitivity	\$10.801M	206.4	0.0005009
Higher Baseline Default	\$10.35M	194.6	0.0010012
Very High Baseline Default	\$9.60M	175.6	0.0020022

Table 5: Impact of default risk

Notes: Parameters used are those in Table 1, apart from b_d which is varied between 100.0 (moderate) and 150.0 (high) to test sensitivity and a_d which is varied between 0.0005 (moderate), 0.001 (higher), and 0.002 (very high) to test baseline default.

As shown in Table 5, the inclusion of default risk leads to a substantial reduction in CMO prices across all scenarios. Unlike prepayment, default results in a direct loss of principal and eliminates future cash flows, making it a more damaging source of risk. The results reflect the fact that increasing default rates significantly reduce CMO price. It is also seen that, as in the prepayment scenarios, baseline default levels have a much larger impact than the sensitivity parameter. In fact, due to the naturally low default rate, tuning the sensitivity parameter has little to no impact on the presented summary statistics. These findings emphasize that credit risk dominates prepayment risk in terms of its effect on valuation.

8.4 Interest Rate Environments

We next analyze the impact of different interest rate environments by varying the initial rate, long-run mean rate, volatility, and mortgage rate.

8.4.1 High vs. Low Rates

Scenario	Price	Avg Life (months)	Prepayment	Default
Low Rates	\$12.20M	199.5	0.00253	0.0005000
High Rates	\$9.51M	211.3	0.00204	0.0005240

Table 6: High vs. low interest rate environment effects

Notes: Parameters used are those in Table 1, apart from r_0 and b . For the low-rate scenario, $r_0 = 0.03$ and $b = 0.03$. For the high-rate scenario, $r_0 = 0.06$ and $b = 0.07$.

As shown in Table 6, low-rate environments produce substantially higher CMO prices than high-rate environments. In low-rate environments, prepayment increases due to refinancing incentives, default decreases, and discounting is reduced. Despite higher prepayment, the reduction in discounting dominates, resulting in higher CMO prices. Conversely, in high-rate environments, prepayment decreases, default increases, and discounting becomes more severe. These effects combine to significantly reduce the value of future cash flows, leading to lower CMO prices. Thus, in this model, higher and lower interest rate levels affect the CMO price more through lenient or aggressive discounting than through prepayment and default changes.

8.4.2 Interest Rate Volatility

We also examine the effect of increasing interest rate volatility.

Volatility	Price	Avg Life (months)	Prepayment	Default
0.03	\$10.80M	206.4	0.00224	0.0005006
0.06	\$10.90M	205.3	0.00228	0.0005069

Table 7: Impact of interest rate volatility

Notes: Parameters used are those in Table 1, apart from σ which is varied between 0.03 and 0.06.

As shown in Table 7, higher interest rate volatility increases both the average prepayment and default rates, as well as the CMO price. Higher volatility introduces more extreme interest rate scenarios. While high-rate scenarios increase default risk, low-rate scenarios increase prepayment and reduce discounting. In the simulations, the beneficial effects of low-rate scenarios slightly outweigh the adverse effects of high-rate scenarios, leading to a modest increase in CMO price.

8.4.3 Mortgage Rate Sensitivity

We now examine the effect of the mortgage rate m , which plays a central role in both prepayment and default behavior.

Mortgage Rate (annual)	Price	Avg Life (months)	Prepayment	Default
5%	\$9.71M	221.9	0.00205	0.0005117
6.5%	\$10.80M	206.4	0.00224	0.0005006
8%	\$11.80M	190.2	0.00247	0.0005000

Table 8: Sensitivity to mortgage rate

Notes: Parameters used are those in Table 1, apart from m which is varied between 0.05/12, 0.065/12, and 0.08/12.

As shown in Table 8, increasing the mortgage rate dramatically increases the price of the CMO. An increase in the mortgage rate leads to higher interest payments, increased prepayment incentives, shorter mortgage lifetimes, and lower default rates. The results indicate that the increase in interest income dominates the effect of accelerated prepayment, resulting in higher CMO prices. This highlights the importance of mortgage rates as a key driver of mortgage-backed security behavior.

8.5 Sensitivity Analysis (Greeks)

To quantify the sensitivity of prices with respect to given parameters, we estimate Greeks using central finite differences with common random seeds. For a parameter θ ,

$$\frac{\partial P}{\partial \theta} \approx \frac{P(\theta + \epsilon) - P(\theta - \epsilon)}{2\epsilon}.$$

Using central finite differences as opposed to forward finite differences reduces bias. In addition, using common random seeds ensures that both evaluations use identical simulated paths, significantly reducing Monte Carlo variance. Alternative approaches such as the likelihood ratio method and pathwise derivative method can provide unbiased estimators. However, these methods require additional analytical structure and can be more difficult to implement, particularly in models with path-dependent features such as stochastic prepayment and default. The finite difference approach provides a simple and robust approximation suitable for this setting.

The sensitivity with respect to the initial interest rate is estimated as

$$\frac{\partial P}{\partial r_0} \approx -2.32 \times 10^7.$$

This corresponds to an approximate change of $-\$23,200$ for a 10 basis point increase in r_0 . This negative relationship is expected and backed up by earlier results. Higher interest rates increase discounting, reducing the present value of future cash flows. Additionally, higher rates increase default intensity in the model, further lowering expected cash flows. These effects dominate any reduction in prepayment activity, resulting in a net decrease in price.

The sensitivity with respect to the interest rate volatility is estimated as

$$\frac{\partial P}{\partial \sigma} \approx 1.87 \times 10^6,$$

implying a price increase of approximately $\$18,700$ for a 1% increase in volatility. This positive sensitivity reflects the asymmetric effects of volatility in the model. Increased volatility raises the likelihood of both low-rate and high-rate scenarios. While high rates increase default risk, low rates increase prepayment activity and reduce discounting. In this model, the beneficial effects of low-rate environments outweigh the adverse effects of high-rate environments, leading to a net positive impact on price.

The sensitivity with respect to the mortgage rate is estimated as

$$\frac{\partial P}{\partial m} \approx 8.42 \times 10^8,$$

which corresponds to a price increase of approximately $\$842,000$ for a 10 basis point increase in m . An increase in the mortgage rate raises the interest component of mortgage payments, increasing cash flows to investors. At the same time, higher m increases the incentive for prepayment in the model, shortening the average mortgage life. The results indicate that the increase in cash flows dominates the effect of accelerated prepayment, resulting in a substantial increase in the CMO price.

The sensitivity with respect to the base prepayment parameter is estimated as

$$\frac{\partial P}{\partial a_p} \approx -1.06 \times 10^8.$$

Higher prepayment intensity reduces the value of the CMO by shortening the life of the mortgage pool and reducing the total interest collected. The sensitivity with respect to the base default parameter is estimated as

$$\frac{\partial P}{\partial a_d} \approx -9.57 \times 10^8.$$

Default has a significantly larger negative impact on price than prepayment. While prepayment returns principal early, default results in a direct loss of principal and future cash flows. The magnitude of this sensitivity highlights that default risk is the dominant source of downside risk in the model. A summary of the computed Greeks and their interpretable effects can be seen in Table 9.

Parameter	Sensitivity	Effect
r_0	-2.32×10^7	-\$23,200 (10 bp effect)
σ	1.87×10^6	+\$18,700 (per 1% increase)
m	8.42×10^8	+\$842,000 (10 bp effect)
a_p	-1.06×10^8	Negative
a_d	-9.57×10^8	Strongly Negative

Table 9: Sensitivity analysis (Greeks)

Notes: Parameters used are those in Table 1, apart from the varied parameter for each Greek. ϵ sizes can be seen within the code in the GitHub repository.

8.6 Summary of Results

The numerical results demonstrate that incorporating increasingly realistic features into the CMO model has a significant impact on valuation. Going from a constant prepayment framework to stochastic prepayment reduces prices by capturing the acceleration of prepayment in favorable rate environments. The introduction of default risk leads to a more substantial decline due to the direct loss of principal and future cash flows. Across both prepayment and default specifications, baseline intensity parameters play a larger role in determining valuation than sensitivity parameters. This highlights the importance of underlying borrower behavior rather than purely rate-driven responses. Interest rate conditions are shown to be a primary driver of CMO pricing, with discounting effects dominating changes in prepayment and default behavior across different rate environments. Higher volatility creates a wider distribution of outcomes, modestly increasing prices as the benefits of low-rate scenarios slightly outweigh the adverse effects of high-rate scenarios. In addition, the mortgage rate itself is a key determinant of value, as higher rates increase cash flows despite accelerating prepayment. Finally, the sensitivity analysis confirms that interest rates, mortgage rates, and default risk are the most influential factors in the model. In particular, default risk can be seen as the dominant source of downside sensitivity, while prepayment mainly affects the timing and magnitude of interest income. Overall, the results illustrate the complex interaction between interest rates, prepayment behavior, and default risk in determining CMO valuation.

9 Conclusion

This paper develops a Monte Carlo framework for pricing collateralized mortgage obligations under stochastic interest rates, with extensions to incorporate both stochastic prepayment and default risk. Starting from a baseline model with constant prepayment, the analysis progressively introduces more realistic features to better capture the behavior of mortgage-backed securities.

The results show that incorporating stochastic prepayment leads to a reduction in CMO prices by accounting for the acceleration of prepayments in low interest rate environments. Extending the model further to include default risk produces a more substantial impact on valuation. Unlike prepayment, which mainly affects the timing of cash flows, default introduces direct losses of principal and future income, making it the dominant source of downside risk in the model. The numerical results also highlight the importance of the interest rate environment. Changes in the level and volatility of interest rates affect valuation through both discounting and borrower behavior, sometimes in competing ways. In particular, the results suggest that discounting effects tend to dominate changes in prepayment and default behavior across different rate scenarios. In addition, the mortgage rate itself plays a key role in determining value, as higher rates increase cash flows despite more prepayment. The sensitivity analysis reinforces these findings by quantifying the relative importance of key parameters. Interest rates and default intensity have the largest negative impact on price, while the mortgage rate has a strong positive effect.

While the model captures several important features of mortgage-backed securities, it is still a simplified framework. The analysis considers a single-tranche structure and does not incorporate factors such as borrower heterogeneity or more detailed recovery assumptions in default. These features could be incorporated in future work to further improve realism. Additionally, more advanced sensitivity estimation techniques, such as likelihood ratio or pathwise derivative methods, could be explored as alternatives to the finite difference approach. Overall, the results illustrate how the interaction between interest rates, prepayment behavior, and default risk drives the valuation of mortgage-backed securities. The Monte Carlo framework provides an intuitive approach for capturing these dynamics, and the extensions developed in this paper demonstrate the importance of moving beyond simplified assumptions when modeling real-world financial instruments.

10 References

1. Buser, Stephen A., and Patric H. Hendershott (1984). The Pricing of Default-Free Fixed Rate Mortgages. *Housing Finance Review*, 3(4), 405–429.
2. Chernov, Mikhail, Francis Longstaff, and Brett R. Dunn (2016). *Macroeconomic-Driven Prepayment Risk and the Valuation of Mortgage-Backed Securities*. CEPR Discussion Paper No. 10947.
3. Deng, Yongheng, John M. Quigley, and Robert Van Order (1999). *Mortgage Terminations, Heterogeneity, and the Exercise of Mortgage Options*. Berkeley Program on Housing and Urban Policy, Working Paper Series qt96r560pg.
4. Dunn, Kenneth B., and John J. McConnell (1981). *A Comparison of Alternative Models for Pricing GNMA Mortgage-Backed Securities*. *The Journal of Finance*, 36(2), 471–484. <https://doi.org/10.1111/j.1540-6261.1981.tb00463.x>
5. Fuster, Andreas, David O. Lucca, and James Ian Vickery (2025). Mortgage-Backed Securities. FRB of Philadelphia Working Paper No. 25-10. Available at SSRN: <https://ssrn.com/abstract=5188947>.
6. Ökten, Giray (2026). Monte Carlo and Quasi-Monte Carlo Methods in Computational Finance. Draft manuscript, Springer Nature.
7. Schwartz, Eduardo S., and Walter N. Torous (1989). *Prepayment and the Valuation of Mortgage-Backed Securities*. *The Journal of Finance*, 44(2), 375–392. <https://doi.org/10.1111/j.1540-6261.1989.tb05062.x>